Mobile App Current Status Report

Disaster Casualty Management System - Progressive Web App and Triage Workflow

Prepared for project review and client discussion

Area	Current Status
PWA readiness	Web export, manifest, service worker, and installable browser experience are implemented. The app can be built to a static **dist** folder for hosting.
Casualty workflow	The Add Casualty form is organized into an 8-step guided workflow covering personal details, incident selection, triage, status, transport, hospital care, and remarks.
Triage logic	The visible triage category input has been removed. The system now derives triage results from assessment answers and backend algorithm logic.
Role filtering	Role-based triage stage visibility is implemented using the account role. Location-aware triage assignment is not yet implemented.

1. Executive Summary

The mobile application has progressed from a React Native mobile interface into a Progressive Web App capable of being exported, hosted, opened in a mobile browser, and installed to a device home screen. The current implementation focuses on field data capture, incident-first casualty logging, triage assessment, and role-sensitive triage stage visibility.

The strongest current improvement is the separation between user-entered triage observations and system-calculated triage results. This supports less biased data collection because overtriage and undertriage summaries are no longer surfaced inside the normal data entry flow.

2. Major Features Implemented

PWA and Deployment Readiness

- The project supports web export through Expo, generating a static **dist** folder that can be served by a static web server or hosted publicly.
- PWA metadata is implemented through a web manifest, service worker, and browser metadata so the app can be installed from supported mobile browsers.
- The app now supports a browser-based mobile workflow, which is useful for testing across devices without requiring app store distribution.

Branding and Landing Page

- The landing page includes institutional logos for UP, UPM DRRMH, DOST, and DOST PCHRD.
- The landing screen preserves the DCMS emergency response identity while introducing the additional partner logos at the top of the page.

Add Casualty Workflow

- The Add Casualty wizard now uses eight steps: Personal, Address, Incident, Triage, Status, Transport, Hospital Care, and Remarks.
- The workflow separates onsite data capture, transport data, and hospital care data more clearly than the earlier single flow.

- The selected incident context is carried into Add Casualty so the user does not repeatedly encode the same disaster information for each casualty.

Reference Data and Quick Create

- Incident quick-create, evacuation center quick-create, and healthcare facility quick-create are collapsible to reduce screen clutter.
- Hazard type is now selected through a searchable drop-down style picker instead of a free text field, reducing spelling variation and improving reporting consistency.
- The hazard list has been expanded to reflect the matrix categories, including geologic, meteorological/hydrological, societal, technological, and biological hazards.

3. Triage Implementation Status

The triage section has been redesigned so the user first identifies the triage stage and triage system, then completes the system-specific assessment popup. The previous visible Triage Category field was removed to avoid depending only on user-selected categories.

Current Triage Data Flow

- The user selects the applicable triage stage based on their role and workflow context.
- The user selects a triage system, such as START, then opens the assessment popup.
- The assessment popup collects the variables needed by the algorithm.
- The backend calculates the triage category based on the submitted assessment answers.
- The system keeps user-entered responder information and calculated triage results available for reporting and verification.

Item	Purpose	Current Behavior
Triage Stage	Identifies when and where the triage decision occurs.	Uses Primary Triage, Secondary Triage, and Tertiary Triage.
Triage System	Identifies the algorithm or framework used for assessment.	Selection opens the assessment workflow for configured systems.
Assessment Popup	Collects algorithm variables.	Shown after a triage system is selected. Final choices are color-coded and arranged evenly.
Calculated Category	Stores the system-derived triage result.	Calculated by backend algorithm logic instead of relying on a visible category field.
Overtriage / Undertriage	Supports reporting and accuracy review.	Hidden from normal mobile data entry to reduce bias, but retained in backend/reporting logic.

4. Role-Based Triage Stage Filtering

Role-based filtering is currently implemented at the account-role level. The app uses the logged-in user's role to decide which triage stages are visible in the Add Casualty triage step.

Role	Visible Triage Stages	Operational Meaning
responder	Primary Triage, Secondary Triage	Designed for field responders or EMT-like users involved at the scene or during reassessment before transfer.
medical_personnel	Tertiary Triage	Designed for facility-based users such as receiving facility personnel, nurses, or hospital staff.
administrator	All stages	Management and oversight role with broader access for operations and correction workflows.
super_admin	All stages	System-level administrative access.

Role	Visible Triage Stages	Operational Meaning
encoder	All stages	Data encoding role that may need to enter records from multiple sources.

5. Workflow and Interface Improvements

- The dashboard Quick Actions are horizontally scrollable, preventing action buttons from being cut off on smaller mobile screens.
- Verification Review has been moved out of individual record details into a dedicated queue, making pending review work easier to find.
- Use current time controls now follow a consistent pattern across time-based fields, improving predictability for users.
- Final triage choices are arranged in a balanced two-by-two layout and color-coded by category so choices are easier to scan.
- Several dense sections were adjusted to improve mobile spacing, reduce visual crowding, and keep content readable on installed PWA screens.

6. Important Limitations

The current role-based triage filtering is based only on account role. The app does not yet know the user's actual reporting location, duty station, assignment, or care setting. For example, it does not yet distinguish whether a user is reporting from the incident scene, an ambulance, the emergency department, a receiving ward, or a command center.

Current Known Gaps

- Location-aware permissions are not yet implemented. A responder assigned to transport and a responder at the scene may still share the same role behavior.
- Role assignment is dependent on Supabase account/profile setup. If the account exists only in the users table and not in Supabase Authentication, login will fail.
- Ngrok-based testing requires separate public URLs for the API and the PWA unless a proper reverse proxy or deployed API is used.
- PWA builds bake environment variables into the exported bundle, so changing the API URL requires rebuilding and re-serving the dist folder.
- The installed PWA may cache an older bundle on iOS; testers may need to close/reopen the app or remove and reinstall the home-screen app after rebuilds.

7. Recommended Improvements

Priority	Improvement	Reason
High	Add reporting-location or assignment context to user sessions.	This would allow triage menus to respond not only to role, but also to whether the user is at scene, transport, ED, or ward.
High	Create a controlled user/account management workflow.	Testing and deployment are easier if admins can create users with correct Auth and profile records from one interface.
Medium	Add a deployment diagnostics screen or API status indicator.	This would help testers see whether the PWA is connected to the correct public API URL.
Medium	Add audit notes for algorithm version and assessment completeness.	This supports later review of why a triage category was calculated and which algorithm rules were applied.
Medium	Improve offline and sync messaging.	Field environments may have unstable connectivity, so users should clearly know whether a record is saved locally, queued, or synced.

Priority	Improvement	Reason
Low	Formalize UI QA checks for installed PWA screens.	Installed iOS PWAs can render spacing differently from desktop browsers, so repeated mobile viewport checks are valuable.

8. Current Status Conclusion

The mobile app is functionally ready for expanded PWA testing and client review of the major workflow changes. The strongest completed areas are the Add Casualty restructuring, disaster-first workflow, collapsible quick-create sections, triage assessment popup, algorithm-based triage calculation, and dashboard queue improvements.

The most important next implementation step is to move from role-only triage filtering to role plus reporting-context filtering. This would better reflect real operational distinctions between bystanders, scene responders, transport personnel, receiving facility nurses, and hospital care teams.